Mandatory Questions and Answers

Question 1: In production, how does your model perform?

I evaluated two approaches: a classical machine learning model using an SVM classifier with CIELAB color histograms and LBP texture features, and a deep learning model using MobileNetV2.

The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Actual production performance depends on the test dataset, image quality, lighting conditions, and hardware.

For production deployment, I would benchmark inference latency on the target hardware and test the models under different lighting conditions, backgrounds, and camera angles before using them in an automated sorting system.

Question 2: What are the drawbacks of your model?

- **Lighting sensitivity:** Changes in illumination, shadows, and reflections can affect the color and texture features used for classification.
- **Limited dataset diversity:** If the training dataset does not include enough variations in lighting, backgrounds, and disease severity, the model may perform poorly on unseen images.
- **Similar visual symptoms:** Different classes may have similar color or texture patterns, making classification difficult.
- **Limited generalization:** Performance on the supplied test dataset may not represent performance on images captured in real-world production environments.

Question 3: If given more time, what would you improve?

- **Improve dataset diversity:** Collect more images across different lighting conditions, camera angles, backgrounds, and disease stages.
- **Enhance model evaluation:** Use cross-validation and evaluate performance separately for each class.
- **Optimize inference speed:** Benchmark both models on the target hardware and investigate ONNX or other suitable deployment optimizations.
- **Improve production robustness:** Add image-quality checks and test the models on images collected directly from a real sorting environment.
- **Explore additional models:** Compare the current SVM and MobileNetV2 approaches with alternatives such as Random Forest or a fine-tuned CNN, using the same data splits and evaluation metrics.

Question 4: Where did you spend the most time while solving your assignment?

I spent most of my time on three areas:

- **Feature engineering:** Extracting CIELAB color histograms and LBP texture features for the SVM model.
- **Model development:** Implementing the classical machine learning pipeline and adapting pretrained MobileNetV2 for the dataset's four classes.

- **Evaluation and comparison:** Preparing the training and validation splits, evaluating the models using accuracy, precision, recall, and F1-score, and generating confusion matrices to understand classification errors.

This process helped me compare a feature-engineering-based approach with a transfer-learning-based deep learning approach.